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1 Basic

1.1 default

```
#include <bits/stdc++.h>
using namespace std;
#define int long long
#define idonthavegirlfriend ios::sync_with_stdio(0),cin
.tie(0);
#define pb push_back
#define endl "\n"
#define all(v) (v).begin(),(v).end()
void solve(){
}
signed main(){
    idonthavegirlfriend
    int _=1;
    //cin>>_;
    while(_--){
        solve();
    }
}
```

1.2 奇妙的東西

```
priority_queue<T> //由大到小
priority_queue<T,vector<T>,greater<T>> //由小到大
__builtin_popcountll // 二進位有幾個1(記得這是long long)
__builtin_clzll // 左起第一個1之前0的個數
__builtin_parityll // 1的個數的奇偶性
__builtin_mul_overflow(a,b,&h) // a*b是否溢位,h = a * b
;
__builtin_add_overflow(a,b,&h)
```

1.3 random

```
mt19937 mt(chrono::steady_clock::now().time_since_epoch
().count());
//mt19937_64 mt() -> return randnum
int randint(int l, int r){
    uniform_int_distribution<> dis(l, r); return dis(mt
);
}
```

2 Binary Search

2.1 二分搜

```
int bsearch_1(int l, int r)
{
    while (l < r)
    {
        int mid = l + r >> 1;
        if (check(mid)) r = mid;
        else l = mid + 1;
    }
    return l;
}
// .....0000000000
int bsearch_2(int l, int r)
{
    while (l < r)
```

```
{
    int mid = l + r + 1 >> 1;
    if (check(mid)) l = mid;
    else r = mid - 1;
}
return l;
}
// 000000000.....
int m = *ranges::partition_point(views::iota(0LL,(int)1
e9+9),[&](int a){
    return check(a) > k;
});
//[begin,last)
//1111111000000000000
//搜左邊數過來第一個 0
//都是 1 會回傳 last
int partitionpoint(int L,int R,function<bool(int)> chk)
{
    int l = L,r = R-1;
    while(r - l > 10){
        int m = l + (r-l)/2;
        if(chk(m)) l = m;
        else r = m;
    }
    int m = l;
    while(m <= r){
        if(!chk(m)) break;
        ++m;
    }
    if(!chk(m)) return m;
    else return R;
}
//手工
```

2.2 三分搜

```
int l = 1,r = 100;
while(l < r) {
    int lmid = l + (r - l) / 3; // l + 1/3區間大小
    int rmid = r - (r - l) / 3; // r - 1/3區間大小
    lans = cal(lmid),rans = cal(rmid);
    // 求凹函數極小直
    if(lans <= rans) r = rmid - 1;
    else l = lmid + 1;
}
```

3 Math

3.1 快速幂

```
int ksm(int x, int y) {
    int ans = 1;
    while (y) {
        if (y & 1)
            ans = ans * x;
        x = x * x;
        y >>= 1;
    }
    return ans;
}
```

// 矩陣乘法

```
vector<vector<int>> mul(const vector<vector<int>> &A,
const vector<vector<int>> &B) {
    if (A.empty() || B.empty() || A[0].empty() || B[0].
empty()) { return {};}
    int m = A.size();
    int k = A[0].size();
    int n = B[0].size();
    vector<vector<int>> C(m, vector<int>(n, 0));
    for (int i = 0; i < m; i++) {
        for (int p = 0; p < k; p++) if (A[i][p]) {
            for (int j = 0; j < n; j++) {
                C[i][j] = (C[i][j] + 1LL * A[i][p] * B[p][j])%MOD;
            }
        }
    }
    return C;
}
```

// 矩陣快速幂

```
vector<vector<int>> ksm(vector<vector<int>> base, int
exp) {
```

```

int n = base.size();
vector<vector<int>> res(n, vector<int>(n, 0));

for (int i = 0; i < n; i++) res[i][i] = 1;

while (exp > 0) {
    if (exp & 1) res = mul(res, base);
    base = mul(base, base);
    exp >>= 1;
}
return res;
}

```

3.2 Prime

```

/* 12721, 13331, 14341, 75577, 123457, 222557, 556679
 * 999983, 1097774749, 1076767633, 100102021, 999997771
 * 1001010013, 1000512343, 987654361, 999991231
 * 999888733, 98789101, 987777733, 999991921, 1010101333
 * 1010102101, 1000000000039, 100000000000037
 * 2305843009213693951, 4611686018427387847
 * 9223372036854775783, 18446744073709551557 */
int mu[ N ], p_tbl[ N ];
vector<int> primes;
void sieve() {
    mu[ 1 ] = p_tbl[ 1 ] = 1;
    for( int i = 2 ; i < N ; i ++ ){
        if( !p_tbl[ i ] ){
            p_tbl[ i ] = i;
            primes.push_back( i );
            mu[ i ] = -1;
        }
        for( int p : primes ){
            int x = i * p;
            if( x >= M ) break;
            p_tbl[ x ] = p;
            mu[ x ] = -mu[ i ];
            if( i % p == 0 ){
                mu[ x ] = 0;
                break;
            }
        }
    }
}
vector<int> factor( int x ){
    vector<int> fac{ 1 };
    while( x > 1 ){
        int fn = SZ(fac), p = p_tbl[ x ], pos = 0;
        while( x % p == 0 ){
            x /= p;
            for( int i = 0 ; i < fn ; i ++ )
                fac.pb( fac[ pos ++ ] * p );
        }
    }
    return fac;
}
}

```

4 題單

4.1 鏈表

```

#include <bits/stdc++.h>
using namespace std;
#define int long long
#define idonhavegirlfriend ios::sync_with_stdio(0),cin.tie(0);
#define pb push_back
#define yn(a) (a?"Yes\n":"No\n")
struct cmp{
    bool operator()(pair<int,int> a,pair<int,int> b)
    {
        if(a.first==b.first) return a.second>b.second;
        return a.first<b.first;
    }
};
signed main(){
    idonhavegirlfriend
    int n;
    cin>>n;
    vector<int> a(n);
    for(auto &c:a)cin>>c;
    vector<pair<pair<int,int>,pair<int,int>>> b;//many
    val pre nxt
    priority_queue<pair<int,int>,vector<pair<int,int>>,
    cmp> q;// many idx
    int cnt=1;
}

```

```

int idx=0;
for(int i=1;i<n;i++){
    if(a[i]==a[i-1]){
        cnt++;
    }
    else{
        b.pb({cnt,a[i-1]},{idx-1,idx+1});
        q.push({cnt,idx});
        cnt=1;
        idx++;
    }
}
b.pb({cnt,a.back()},{idx-1,idx+1});
q.push({cnt,idx});
cnt=1;
idx++;
int ans=0;
vector<bool> vis(b.size(),false);
while(!q.empty()){
    auto [x,id]=q.top();
    q.pop();
    if(vis[id]) continue;
    ans++;
    vis[id]=true;
    auto [pp,pn]=b[id].second;
    if(pp==-1 and pn==b.size()) continue;
    if (pp!=-1) b[pp].second.second=pn;
    if (pn!=b.size()) b[pn].second.first=pp;
    if(pn==b.size() and pp==-1) continue;
    if(pn!=b.size() and pp!=-1){
        if(b[pn].first.second==b[pp].first.second){
            int cnt=b[pn].first.first+b[pp].first.first;
            vis[pn]=true;
            b[pp].first.first=cnt;
            b[pp].second.second=b[pn].second.second;
        }
        if(pn!=b.size()){
            int pnn=b[pn].second.second;
            if(pnn!=b.size()){
                b[pnn].second.first=pp;
            }
        }
        q.push({cnt,pp});
    }
}
cout<<ans;
}

```



— *

[illegible]